

## Project Planning Phase

### Product Backlog, Sprint Schedule, and Estimation

| Sprint No. | Duration (Weeks) | Start Date | End Date   | Goals / Deliverables                                       | Estimation (Person-Days) | Notes                                    |
|------------|------------------|------------|------------|------------------------------------------------------------|--------------------------|------------------------------------------|
| Sprint 1   | 2                | 2026-06-13 | 2026-06-14 | Requirement analysis, environment setup, data collection   | 3                        | Setup network data capture tools         |
| Sprint 2   | 2                | 2026-06-14 | 2026-06-15 | Data preprocessing pipeline, feature engineering           | 3                        | Develop scripts for feature extraction   |
| Sprint 3   | 3                | 2026-06-16 | 2026-06-17 | Initial AI model development (classical ML baseline)       | 2                        | Train/test Random Forest, SVM            |
| Sprint 4   | 3                | 2026-06-16 | 2026-06-16 | Deep learning model design and prototyping                 | 5                        | LSTM/CNN models for anomaly detection    |
| Sprint 5   | 2                | 2026-06-17 | 2026-06-17 | Integration of detection engine with real-time data stream | 6                        | Connect models with data pipeline        |
| Sprint 6   | 2                | 2026-06-18 | 2026-06-18 | Alerting system and dashboard UI prototype                 | 4                        | Setup notification and visualization     |
| Sprint 7   | 2                | 2025-07-23 | 2025-08-05 | System testing, performance tuning, and documentation      | 3                        | Test for false positives, optimize speed |
| Sprint 8   | 1                | 2026-06-19 | 2026-06-19 | Final deployment and user training                         | 5                        | Deploy system and train users            |

